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CSE 13S Spring 2021
Assignment 1: Left, Right and Center
Design Document

In the game *Left, Right, and Center*, there are 2 to 14 players that each have $3 and three dice with 6 faces each. 3 of the faces are a •, 1 is an L (left), 1 is an R (right), and 1 is a C (center). Player 1 rolls 3 dice if he or she has $3 or more. A player rolls 2 dice if he or has $2 and 1 die if he or she has $1. Players pass if they have no money.
For each die:

- If the player rolls L, he gives $1 to the player on his left.
- If the player rolls R, he gives $1 to the player on his right.
- If the player rolls C, he puts the $1 in the pot in the center.
- If the player rolls •, he ignores it.

Then move to the next player to the right. Repeat until only one player has any money. That remaining player wins the pot in the center.

I will implement this game by using the random() function to simulate rolling a dice. The range will be from 0-5 to represent the six faces of the dice.

The inputs to our program are:

1. The random seed
2. The number of players (must be from 1 to 14)

Expected outputs:

1. Error message if the seed is invalid (must be a number greater than 1)
2. Error message if the number of players is invalid (must be a number from 1 to 14)
3. A correct simulation of the game if the random seed and number of player inputs are valid

When using the same random seed and the same number of players, the game simulation will always be the same.

TOP LEVEL:

The top level design of the code is given by the following pseudocode:

Left function:
    Return ((position + players - 1) % players)

Right function:
    Return ((position + 1) % players)

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Roll function:
    Return (random number % n)

Main:
    Enumerate pass, left, right, center with type faces
    die array of type faces = { LEFT, RIGHT, CENTER, PASS, PASS }
    dollars array = { 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3 }
    Declare integer seed
    Declare integer number of players
    Declare integer roll state
    Pot = 0

    Print “Random seed: “
    Read the seed
    If pointer to seed < 1:
        Print “Pseudorandom seed must be non-negative (<seed>)”
        End program
        Print new line
    Print “How many players? “
    Read the number of players
    If the pointer to the number of players < 1 or number of players  is not > 1 and <= 14:
        Print(“Number of players must be from 1 to 14.”)
        Print new line
        End program
    Set random seed

    Alive = number of players
    While alive > 1:

        For each player:
            Roll state = 0
            Alive = 0
            For each player:
                If player’s dollars > 0
                    Increment alive
            If alive == 1:
                Break
            If player’s dollars > 0:
                Print “<philosopher> rolls…”
                Roll state = 1
            Current dollars = player’s dollars
            Declare integer min
            Declare integer space count
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                If 3 < current dollars:
                        Min = 3
                Else:
                        Min = current dollars

                For min times:
                        Roll num = roll(6)
                        If player rolled LEFT:
                                Decrement player's dollars
                                Increment left player's dollars
                                Print "gives $1 to <left philosopher>"

                        If player rolled RIGHT:
                                Decrement player's dollars
                                Increment right player's dollars
                                Print "gives $1 to <right philosopher>"

                        If player rolled CENTER:
                                Decrement player's dollars
                                Increment pot
                                Print "puts $1 in the pot"

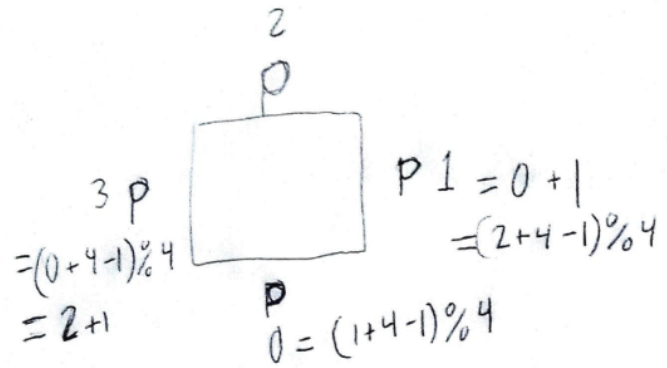
                        If player rolled PASS
                                Print "gets a pass"

                        If space count < min - 1:
                                Print " "

                If roll state == 1:
                        Print new line

For each player:
        If player's dollars > 0:
                Print "<philosopher> wins the $<pot> pot with $<player's dollars> left in
                the bank"
                Print new line
End program
```

PAPER DRAFT AND WORK



P0 roll x3

P1 roll x3

P2 roll x3

P3 roll x3

num rolls = num $

at start: num alive = num players

if player x has $0, num alive -=1

if player with $0 is given $1, num alive +=1

if num alive = 1, end game